HARDWARE DOCUMENT

VNT-001

First edition
2026-10-25

Vantage — Sensor board — Fabrication, assembly and test

Applies to: Vantage sensor board, Rev A (Vantage_Sensor.kicad_pcb, Rev A)

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Foreword

Team sevenUP prepares and maintains the hardware documents of the Vantage sensor node. The design files, the drawings and these documents are kept together in the team hardware repository, so that every hardware revision has a matching set of documents.

This document was prepared by Team sevenUP.

This first edition applies to Rev A of the sensor board. A new edition is issued whenever the sensor board revision changes. The changes between revisions are recorded in the repository change log (`CHANGELOG.md`).

Questions and proposed changes to this document should be raised as issues in the hardware repository.

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Introduction

The Vantage sensor node is the field unit of a flood-monitoring system. It measures water level, rainfall and ambient conditions and sends the data over Wi-Fi. The node is built from two boards connected by one cable:

— the **sensor board**, which carries an ESP32-S3 development board and the interfaces to the sensors;

— the **power board**, which carries the battery, the solar charger, the regulators and the power monitoring (see VNT-002).

The sensor board is designed to be made by hand with the toner-transfer method. It therefore has a single copper layer, wide tracks and gaps, large pads and hand-solderable parts. Two short wire links on the component side complete the ground connections that a single copper layer cannot make.

This document gives the procedure to make the board, to assemble it, to configure it and to check that it works. The design files themselves are listed in Clause 2.

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1 Scope

This document specifies the fabrication, assembly, configuration, inspection and testing of the Vantage sensor board, Rev A.

It applies to boards made by toner transfer and wet etching on single-sided copper laminate. It does not cover the power board (see VNT-002), the firmware, or the mechanical enclosure, except where the sensor board places requirements on them.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For each, the Rev A file in the hardware repository applies.

— `sensor-board/Vantage_Sensor.kicad_sch`, *Vantage — Sensor board — Schematic*

— `sensor-board/Vantage_Sensor.kicad_pcb`, *Vantage — Sensor board — PCB layout*

— `outputs/sensor-board/Vantage_Sensor_Copper-Artwork_Letter.pdf` and `…_A4.pdf`, *Vantage — Sensor board — Copper artwork, 1:1*

— `outputs/sensor-board/Vantage_Sensor_Assembly-Drawing.pdf`, *Vantage — Sensor board — Assembly drawing*

— `outputs/sensor-board/Vantage_Sensor_BOM.csv`, *Vantage — Sensor board — Bill of materials*

— VNT-002, *Vantage — Power board — Fabrication, assembly and test*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
component side

side of the board on which the through-hole components and the development board are placed

Note 1 to entry: The component side carries no copper pattern, only the two wire links.

3.2
copper side

side of the board that carries the copper pattern; the surface-mount components are soldered on this side

3.3
copper artwork

1:1 printed image of the copper pattern, used as the etch resist in toner transfer

3.4
toner transfer

method of making a printed circuit board in which laser-printer toner is transferred onto copper laminate by heat and pressure and then serves as the etch resist

3.5
wire link

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short piece of bare tinned wire on the component side that joins two copper areas which the single copper layer cannot join

3.6
jumper

pair of header pins that is closed by a removable jumper cap

3.7
dev board

ESP32-S3 development board (YD-ESP32-S3, ESP32-S3-DevKitC-1 compatible, N16R8) that plugs into the sockets of U1

3.8
bench mode

configuration in which the sensor board is powered from the USB port of the dev board only

3.9
field mode

configuration in which the sensor board is powered by the power board through J1

3.10
sensor 5 V

5 V supply rail (net V5_SENS) for the ultrasonic module and the TF-Luna

3.11
SENS_EN

control signal from GPIO21 of the dev board that switches the sensor 5 V on the power board; high level = on

4 Abbreviated terms

Abbreviation	Meaning
GPIO	General-purpose input/output
I2C	Inter-integrated circuit (two-wire serial bus)
PTC	Positive temperature coefficient (resettable) fuse
SMD	Surface-mount device
THT	Through-hole technology
TVS	Transient voltage suppressor (diode)
UART	Universal asynchronous receiver-transmitter (serial port)

5 Description

5.1 General

The sensor board is a carrier for the dev board. It distributes the supplies, protects the sensor lines and connects the sensors to the GPIOs of the dev board. Its main characteristics are given in Table 1. The component side is shown in Figure 1 and the copper side in Figure 2.

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Table 1 — Main characteristics

Characteristic	Value
Outline	110.0 mm × 30.0 mm, rectangular
Base material	Single-sided FR-4 copper laminate, 1.6 mm thick, 35 µm copper (recommended)
Copper	One layer on the copper side, with a ground fill and thermal reliefs
Track width	0.6 mm for signals, 1.0 mm for supplies and ground
Minimum clearance	0.5 mm copper to copper, 0.8 mm copper to board edge
Holes	72 × 1.0 mm, 2 × 1.2 mm, 2 × 3.2 mm (see Annex B)
Wire links	2 (L1, L2), on the component side
Surface-mount parts	16, all on the copper side (1206, 1812 and SMA packages)
Through-hole parts	Dev board sockets (U1), J1 to J5, JP1, JP2
Mounting	2 × M3: H1 under the dev board, H2 at the J1 end
Antenna area	No ground fill under the antenna end of the dev board

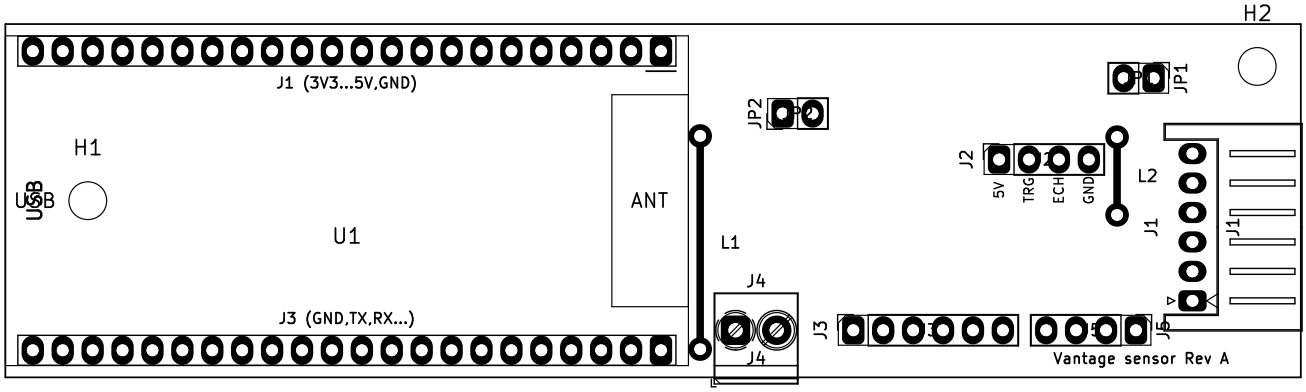


Figure 1 — Component side (USB end on the left, J1 end on the right)

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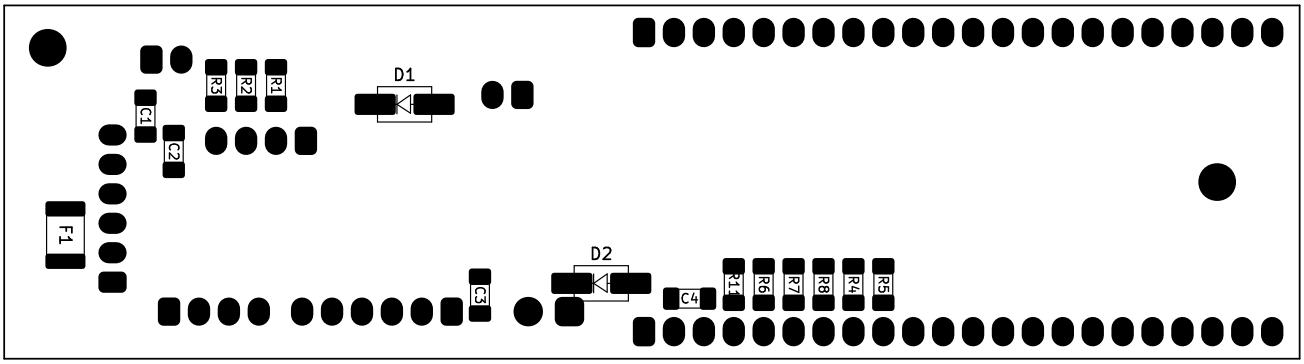


Figure 2 — Copper side, seen from the copper side, showing the pads and the surface-mount parts

5.2 Interfaces

The connectors and jumpers are listed in Table 2. The pin assignments are given in Annex C.

Table 2 — Connectors and jumpers

Ref.	Function	Type
U1	Dev board	2 × 22-way female header, 2.54 mm, rows 25.4 mm apart
J1	Power board connection: 3.3 V, sensor 5 V, ground, I2C, SENS_EN	JST XH, 6-way, side entry
J2	AJ-SR04M ultrasonic module (stands upright)	4-way female header, 2.54 mm
J3	BME280 module (6-pin module)	6-way female header, 2.54 mm
J4	Rain gauge (reed switch)	2-way screw terminal, 3.5 mm
J5	TF-Luna LiDAR (optional)	4-way male header, 2.54 mm
JP1	3V3 FEED: power-board 3.3 V to the dev board	2-way male header + jumper cap
JP2	BENCH 5V: dev-board USB 5 V to the sensor 5 V	2-way male header + jumper cap

5.3 Protection functions

The sensor lines leave the enclosure and are protected as follows:

— the rain-gauge input has a TVS diode (D2, SMAJ6.0A), a 1 kΩ series resistor (R11) and a 100 nF filter capacitor (C4);

— the BME280 is supplied from GPIO2 through 100 Ω (R6), so that a short circuit in its cable is limited and the sensor can be switched off; its I2C lines have 220 Ω series resistors (R7, R8);

— the ultrasonic trigger and the TF-Luna UART lines have 1 kΩ series resistors (R1, R4, R5); the echo line has a divider (R2 10 kΩ, R3 18 kΩ) that reduces the 5 V echo to about 3.2 V;

— the TF-Luna supply has a resettable fuse (F1, 0.2 A);

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— the bench 5 V passes through a Schottky diode (D1, SS14) so that the sensor 5 V can never feed back into the USB port.

6 Safety

WARNING — Etching solutions such as ferric chloride and sodium persulfate are corrosive and stain skin, clothing and surfaces. Wear gloves and eye protection, work in a ventilated place, and dispose of used etchant according to local rules. Do not pour etchant into a drain.

WARNING — Acetone and isopropyl alcohol are flammable. Keep them away from the iron and the soldering iron.

WARNING — Wear eye protection when drilling and cutting, and clamp the board. Ventilate when soldering and wash hands after handling solder.

7 Materials and tools

The parts are listed in Annex A. The materials and tools in Table 3 are also needed.

Table 3 — Materials and tools

Item	Requirement
Copper laminate	Single-sided FR-4, at least 114 mm × 34 mm
Printer	Laser printer (inkjet prints do not act as etch resist)
Transfer medium	Toner-transfer paper or glossy magazine paper
Heat source	Clothes iron without steam, or a laminator
Etchant	Ferric chloride, or sodium persulfate, with a plastic tray
Cleaning	Fine steel wool or 800-grit abrasive paper, isopropyl alcohol, acetone
Touch-up	Permanent marker suitable as etch resist
Drilling	Drill press or rotary tool on a stand; bits 1.0 mm, 1.2 mm and 3.2 mm
Cutting	Hacksaw or board shear, flat file
Soldering	Temperature-controlled iron with a fine tip, solder 0.5 mm to 0.8 mm, flux, desoldering braid
Wire	Bare tinned copper wire, about 0.6 mm, for the wire links
Measuring	Ruler or calliper, digital multimeter

8 Fabrication of the printed circuit board

8.1 Printing the copper artwork

The copper artwork shall be printed from `Vantage_Sensor_Copper-Artwork_Letter.pdf` (US Letter) or `Vantage_Sensor_Copper-Artwork_A4.pdf` (A4), whichever matches the paper in use. Each sheet carries four copies of the board, each with a 100 mm scale line and drill-centre marks.

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The procedure shall be as follows.

a) In the print dialogue, select 100 % or “Actual size”. Disable “Fit to page”, “Shrink oversized pages” and any other scaling. Set the paper size in the printer properties to the size of the sheet.

b) Select the darkest print setting. Do not mirror the print. The artwork is already the right way round for transfer; the copper text reads backwards on the paper.

c) Print one sheet on plain paper. Measure a scale line and a board outline.

d) Accept the print only if the scale line measures 100.0 mm ± 0.5 mm and the board outline measures 110.0 mm × 30.0 mm ± 0.5 mm. Otherwise correct the printer settings and repeat from a).

e) Print the accepted settings on the transfer medium.

NOTE A scale line of about 97 mm means that the page was shrunk to fit another paper size, typically a Letter page printed with the printer set to A4. Correct the paper size setting instead of scaling the artwork, because the header rows of the dev board need the full 1:1 size.

8.2 Preparing the laminate

Cut a blank of at least 114 mm × 34 mm, so that the board outline has a margin of at least 2 mm on each side. Remove the burrs from the edges. Rub the copper with fine steel wool or abrasive paper until it is evenly bright, then clean it with isopropyl alcohol. Do not touch the cleaned copper.

8.3 Transferring the toner

a) Cut out one copy of the artwork with a margin of about 5 mm and place it toner side down on the copper.

b) Apply heat and firm pressure with the iron, without steam, at the cotton setting, for 3 min to 5 min, moving the iron so that the whole copy is heated evenly. With a laminator, pass the board through 8 to 10 times.

c) Let the board cool, soak it in warm water until the paper is soft, and rub the paper off.

d) Inspect the toner pattern against the artwork. All tracks and pads shall be complete and all gaps shall be open. Small defects may be repaired with the etch-resist marker; toner in a gap may be removed with a needle. If more than a few defects are found, clean the copper and repeat the transfer.

8.4 Etching

Etch the board following the instructions of the etchant supplier, moving the tray gently. Check the board often near the end of the etch and remove it as soon as all exposed copper is gone. Rinse it in clean water.

8.5 Removing the toner and inspecting the copper

Remove the toner with acetone. Inspect the copper under good light, with a magnifier if available, against the artwork. There shall be no copper bridge between tracks or pads and no break in a track.

8.6 Drilling

Drill the holes from the copper side at the drill-centre marks, with the sizes given in Table B.1. Drill the 3.2 mm holes with a 1.0 mm pilot hole first. H2 is only 2 mm from the board edge; drill it slowly. Remove the burrs after drilling.

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8.7 Cutting to size

Cut the board about 1 mm outside the outline, then file it to the outline. The finished board shall measure 110.0 mm × 30.0 mm ± 0.3 mm. The copper is 0.8 mm from the edge; do not file into it.

8.8 Protecting the copper

The copper should be protected against oxidation before assembly, for example by tinning it or by applying a thin coat of rosin flux dissolved in isopropyl alcohol.

9 Assembly

9.1 General

Through-hole parts are placed on the component side and soldered on the copper side. Surface-mount parts are placed and soldered on the copper side. The positions are shown in Figures 1 and 2 and in the assembly drawing.

The parts shall be fitted in the order of Table 4, so that each part can still be reached for soldering.

Table 4 — Order of assembly

Step	Part	Side	Remarks
1	Wire links L1 and L2	Component	Bare tinned wire, flat on the board. L2 runs beside J4 and is fitted before J4. Positions in Table B.2.
2	R1 to R8, R11	Copper	1206. Values in Annex A.
3	C1 to C4	Copper	1206.
4	F1	Copper	1812.
5	D1, D2	Copper	SMA. Polarity in Table 5.
6	JP1, JP2	Component	2-way headers.
7	J5	Component	4-way male header.
8	J3, J2	Component	Female headers. Keep them square to the board.
9	J4	Component	Wire entry towards the long board edge.
10	J1	Component	Opening towards the J1 end of the board.
11	U1 sockets	Component	2 × 22-way. Plug the dev board in while soldering to keep the rows aligned.
12	Mounting at H1	—	H1 is under the dev board. Fit its screw or standoff before the dev board.

9.2 Orientation

The parts in Table 5 shall be fitted in the orientation given. On every connector, pin 1 has a square pad.

Table 5 — Orientation of polarized parts

Part	Orientation
D1 (SS14)	Cathode band towards the J1 end of the board
D2 (SMAJ6.0A)	Cathode band towards the J1 end of the board

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Part	Orientation
J1	Opening of the housing towards the J1 end; the cable leaves past the board end
J4	Wire entry towards the long edge next to it
Dev board	USB connectors at the USB end (H1 end) of the board
AJ-SR04M in J2	Pin 1 (5V) of the module in the square pad of J2
BME280 in J3	Pin 1 (VCC) of the module in the square pad of J3

9.3 Soldering

For each surface-mount part, tin one pad, place the part, reflow that pad, then solder the other pad. Check that no solder bridges two pads.

For each through-hole part, insert it fully, check that it sits flat, solder one pin, check the alignment, then solder the remaining pins.

9.4 Dev board

The board accepts dev boards whose header rows are 25.4 mm apart, such as the YD-ESP32-S3. Before buying a replacement, check the row spacing; the ESP32-S3-DevKitC-1 made by Espressif is narrower and does not fit.

For field mode on battery, the power LED of the dev board should be removed (or its series resistor), because it draws current whenever the board has 3.3 V.

10 Configuration

10.1 Operating modes

The jumpers shall be set according to Table 6.

Table 6 — Jumper settings

Mode	JP1 (3V3 FEED)	JP2 (BENCH 5V)	Power board
Bench mode	Open	Fitted	Not connected
Field mode	Fitted	Open	Connected through J1
Current measurement	Open, ammeter across the two pins	Open	Connected through J1

JP2 shall be open whenever the power board is connected. Otherwise the USB supply keeps the sensor 5 V on when SENS_EN switches it off.

The USB port of the dev board may be connected in field mode, for example for programming.

NOTE For a node installed outdoors, JP1 may be replaced by a soldered wire. A jumper cap can work loose with vibration or corrode, and an open JP1 removes the supply of the dev board.

10.2 Dev board 5 V output for bench mode

In bench mode the sensor 5 V comes from the 5V pin of the dev board. On the YD-ESP32-S3 a diode separates this pin from the USB supply, and the pin gives no output until the solder pads marked IN-OUT on the dev board are bridged.

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With the dev board on USB, measure the 5V pin against GND. If it reads less than 4.5 V, bridge the IN-OUT pads. The sensor board cannot feed voltage back into this pin, because D1 blocks it.

10.3 Cable to the power board

The cable between J1 of the sensor board and J1 of the power board shall be a 6-way JST XH cable wired pin 1 to pin 1. It should be no longer than 0.3 m, because the I2C bus runs through it. Before the first connection, check with a multimeter that each pin connects to the same pin number at the other end and that no two wires are connected.

11 Inspection and testing

11.1 General

The tests shall be made in the order of this clause. A board is accepted when every test passes. The results should be recorded on the form in Annex E.

11.2 Visual inspection

Check that all parts are fitted in the right position and orientation (Tables 4 and 5), that all joints are shiny and fully wetted, and that no solder bridge, flux residue or loose wire remains.

11.3 Resistance checks

Make the checks in Table 7 with the dev board and all modules removed. “U1 IOn” means the socket contact for dev board pin IOn (Annex C).

Table 7 — Resistance checks

No.	Between	Expected
R-1	J1-5 (GND) and J2-4, J3-2, J4-2, J5-4, U1 GND (J3-1) in turn	Below 1 Ω (checks L1 and L2)
R-2	J1-6 and J1-5	Above 100 kΩ after C1 has charged
R-3	J1-4 and J1-5	Above 100 kΩ after C2 has charged
R-4	JP1-2 and J1-5	Above 100 kΩ
R-5	J2-3 and J1-5	28 kΩ ± 5 % (R2 + R3)
R-6	J2-3 and U1 IO4	10 kΩ ± 5 % (R2)
R-7	J2-2 and U1 IO5	1 kΩ ± 5 % (R1)
R-8	J3-1 and U1 IO2	100 Ω ± 5 % (R6)
R-9	J3-3 and U1 IO42; J3-4 and U1 IO41	220 Ω ± 5 % each (R7, R8)
R-10	J4-1 and U1 IO1	1 kΩ ± 5 % (R11)
R-11	J5-3 and U1 IO40; J5-2 and U1 IO39	1 kΩ ± 5 % each (R4, R5)
R-12	J5-1 and J1-4	Below 10 Ω (F1)
R-13	JP2-1 and U1 5V	Below 1 Ω
R-14	D1, diode test, red lead on JP2-2, black lead on J1-4	0.15 V to 0.45 V; open with the leads reversed

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11.4 Bench power-up test

Set bench mode (Table 6), plug in the dev board without modules and connect its USB port to a supply of at least 1 A. The voltages in Table 8 shall be within the limits given.

Table 8 — Bench-mode voltages

No.	Point	Condition	Limits
V-1	JP1-2 to GND	—	3.20 V to 3.40 V
V-2	JP2-1 to GND	IN-OUT bridged (10.2)	4.60 V to 5.25 V
V-3	J2-1 to GND	IN-OUT bridged (10.2)	4.30 V to 5.10 V
V-4	J3-1 to GND	GPIO2 set high by firmware	3.10 V to 3.40 V

11.5 Functional tests

Fit the modules and load a test firmware that follows Annex D. Each interface shall work as given in Table 9.

Table 9 — Functional tests

No.	Interface	Test	Pass condition
F-1	BME280 (J3)	I2C scan on GPIO41/GPIO42 with GPIO2 high	Device found at 0x76 or 0x77; plausible temperature, humidity and pressure
F-2	Ultrasonic (J2)	Measure a flat target at a taped distance between 0.3 m and 2 m	Reading within ± 2 cm of the taped distance
F-3	Rain gauge (J4)	Tip the bucket (or short J4-1 to J4-2) ten times	Exactly ten pulses counted
F-4	TF-Luna (J5), if fitted	Read the UART at 115 200 baud	Frames starting with 0x59 0x59 received; distance plausible

11.6 Field-mode test

After the power board has passed the tests of VNT-002, set field mode (Table 6) and connect the power board. The results shall be as given in Table 10.

Table 10 — Field-mode tests

No.	Test	Pass condition
P-1	Voltage JP1-2 to GND	3.20 V to 3.40 V
P-2	I2C scan on GPIO38/GPIO47	Devices found at 0x40 and 0x44
P-3	J2-1 to GND with SENS_EN low	Below 0.3 V
P-4	J2-1 to GND with SENS_EN high	4.75 V to 5.25 V
P-5	Repeat F-1 to F-4 in field mode	As in Table 9

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Annex A
(normative)
Bill of materials

Table A.1 lists the parts fitted to the board. Table A.2 lists the other items needed to complete a node. `Vantage_Sensor_BOM.csv` holds the same list in machine-readable form.

Table A.1 — Parts fitted to the board

Item	Qty	Reference	Value	Package	Side
1	1	C1	10 µF	1206, X7R or X5R, ≥ 10 V	Copper
2	1	C2	22 µF	1206, X7R or X5R, ≥ 10 V	Copper
3	1	C3	1 µF	1206, X7R or X5R, ≥ 10 V	Copper
4	1	C4	100 nF	1206, X7R or X5R, ≥ 10 V	Copper
5	1	D1	SS14	SMA (DO-214AC)	Copper
6	1	D2	SMAJ6.0A	SMA (DO-214AC)	Copper
7	1	F1	PTC 0.2 A	1812	Copper
8	1	J1	POWER	JST S6B-XH-A, XH 6-way side entry	Component
9	1	J2	AJ-SR04M	1 × 4 female header, 2.54 mm	Component
10	1	J3	BME280	1 × 6 female header, 2.54 mm	Component
11	1	J4	RAIN	2-way screw terminal, 3.5 mm pitch	Component
12	1	J5	TF-Luna	1 × 4 male header, 2.54 mm	Component
13	1	JP1	3V3 FEED	1 × 2 male header, 2.54 mm, with jumper cap	Component
14	1	JP2	BENCH 5V	1 × 2 male header, 2.54 mm, with jumper cap	Component
15	4	R1, R4, R5, R11	1 kΩ	1206	Copper
16	1	R2	10 kΩ	1206	Copper
17	1	R3	18 kΩ	1206	Copper
18	1	R6	100 Ω	1206	Copper
19	2	R7, R8	220 Ω	1206	Copper
20	1	U1	Dev board sockets	2 × (1 × 22 female header, 2.54 mm), rows 25.4 mm apart	Component

Table A.2 — Other items for one node

Item	Qty	Description
1	1	Dev board YD-ESP32-S3 N16R8 (ESP32-S3-DevKitC-1 compatible), header rows 25.4 mm apart
2	1	AJ-SR04M ultrasonic module with waterproof transducer
3	1	BME280 module, 6-pin (VCC, GND, SCL, SDA, CSB, SDO)
4	1	Rain gauge with reed switch (WH-SP-RG type)
5	1	TF-Luna LiDAR module (optional)

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Item	Qty	Description
6	1	Cable, JST XH 6-way, pin 1 to pin 1, 0.3 m maximum
7	2	M3 screw with standoff
8	—	Bare tinned copper wire, about 0.6 mm, for L1 and L2

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Annex B
(normative)
Drilling and wire links

B.1 Drill sizes

Table B.1 — Drill sizes

Drill	Use	Count
1.0 mm	Component pins: headers, sockets, J1, jumpers	68
1.0 mm	Wire links L1 and L2 (two holes each)	4
1.2 mm	Screw terminal J4	2
3.2 mm	Mounting holes H1, H2 (no copper)	2

B.2 Wire links

The coordinates in Table B.2 are measured on the component side from the corner at the USB end and the J1-header edge of the dev board (top left in Figure 1), X along the board and Y across it.

Table B.2 — Wire links

Link	From (X, Y)	To (X, Y)	Length	Function
L1	94.4 mm, 9.6 mm	94.4 mm, 16.2 mm	6.6 mm	Carries ground from the J1 and J2 area over the supply and echo tracks
L2	59.0 mm, 9.5 mm	59.0 mm, 27.6 mm	18.1 mm	Carries ground over the dev board signal tracks to the J3, J4 and J5 edge

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Annex C
(informative)
Connector pin assignments

Table C.1 — J1, power board connection

Pin	Signal	Dev board pin	Remarks
1	I2C_SDA	IO38	I2C to the INA226 monitors on the power board; pull-ups on the power board
2	I2C_SCL	IO47	
3	SENS_EN	IO21	High = sensor 5 V on; pulled low on the power board
4	Sensor 5 V	—	Switched 5 V from the power board
5	GND	GND	
6	3.3 V	3V3 through JP1	Supply of the dev board in field mode

Table C.2 — J2, AJ-SR04M ultrasonic module

Pin	Signal	Connection
1	5V	Sensor 5 V
2	Trig	IO5 through R1 (1 kΩ)
3	Echo	IO4 through R2 (10 kΩ), R3 (18 kΩ) to GND
4	GND	GND

Table C.3 — J3, BME280 module

Pin	Signal	Connection
1	VCC	IO2 through R6 (100 Ω); C3 (1 µF) to GND
2	GND	GND
3	SCL	IO42 through R7 (220 Ω)
4	SDA	IO41 through R8 (220 Ω)
5	CSB	Not connected (set by the module)
6	SDO	Not connected (set by the module)

Table C.4 — J4, rain gauge and J5, TF-Luna

Pin	J4 (rain gauge)	J5 (TF-Luna)
1	Switch: IO1 through R11 (1 kΩ); D2 and C4	5V: sensor 5 V through F1 (0.2 A)
2	Switch: GND	RXD: IO39 through R5 (1 kΩ)
3	—	TXD: IO40 through R4 (1 kΩ)
4	—	GND

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Table C.5 — Dev board pins used

Dev board pin	Signal	Direction	Use
3V3 (both)	3.3 V	Supply	From JP1 in field mode
5V	USB 5 V	Supply	To JP2 (bench mode)
GND (J3 row, pin 1)	GND	—	Ground
IO1	RAIN_PULSE	Input	Rain gauge (RTC pin, wake from deep sleep)
IO2	BME_PWR	Output	BME280 supply
IO4	US_ECHO	Input	Ultrasonic echo
IO5	US_TRIG	Output	Ultrasonic trigger
IO21	SENS_EN	Output	Sensor 5 V on/off
IO38	I2C_SDA	Bidirectional	Power board I2C
IO39	LID_TX	Output	TF-Luna UART, to its RXD
IO40	LID_RX	Input	TF-Luna UART, from its TXD
IO41	BME_SDA	Bidirectional	BME280 I2C
IO42	BME_SCL	Output	BME280 I2C
IO47	I2C_SCL	Output	Power board I2C

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Annex D

(informative)

Firmware requirements related to the hardware

The firmware should observe the following rules, which follow from the circuit.

a) Keep SENS_EN (IO21) low except while the sensors are measuring. Hold it low through deep sleep. The power board pull-down keeps the sensor 5 V off while the pin is not driven.

b) While the sensor 5 V is off, drive US_TRIG (IO5) low and drive LID_TX (IO39) low or set it as an input. A UART output idles high and would otherwise feed current into the unpowered TF-Luna through R5.

c) While the BME280 supply (IO2) is low, set IO41 and IO42 as inputs without pull-ups, so that the I2C lines do not feed the unpowered sensor.

d) Use the internal pull-up on IO1 for the rain gauge, also in deep sleep, and wake on the falling edge. Debounce in software for at least 10 ms.

e) Probe the BME280 at address 0x76 and 0x77; the address depends on the module.

f) The INA226 monitors are at addresses 0x40 (solar) and 0x44 (battery) on IO38/IO47. Without the power board the bus has no pull-ups; enable the internal pull-ups or do not use the bus.

g) Do not use the strapping pins (IO0, IO3, IO45, IO46), the PSRAM pins (IO35 to IO37), the USB pins (IO19, IO20) or the RGB LED pin (IO48) for other functions.

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Annex E
(informative)
Test record

Board serial number	Date	Tested by

Table E.1 — Test record

Test	Clause	Measured value	Pass / fail
Visual inspection	11.2		
R-1 to R-14	11.3		
V-1	11.4		
V-2	11.4		
V-3	11.4		
V-4	11.4		
F-1 BME280	11.5		
F-2 Ultrasonic	11.5		
F-3 Rain gauge	11.5		
F-4 TF-Luna	11.5		
P-1 to P-5	11.6		
Result: accepted / rejected			

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d) Bosch Sensortec, *BME280 Combined humidity and pressure sensor — Datasheet*

e) Benewake, *TF-Luna LiDAR Module — Product Manual*

f) *AJ-SR04M ultrasonic ranging module — Module documentation*

g) JST, *XH Connector — Catalogue*

h) ISO/IEC Directives, Part 2, *Principles and rules for the structure and drafting of ISO and IEC documents* (layout model of this document)

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